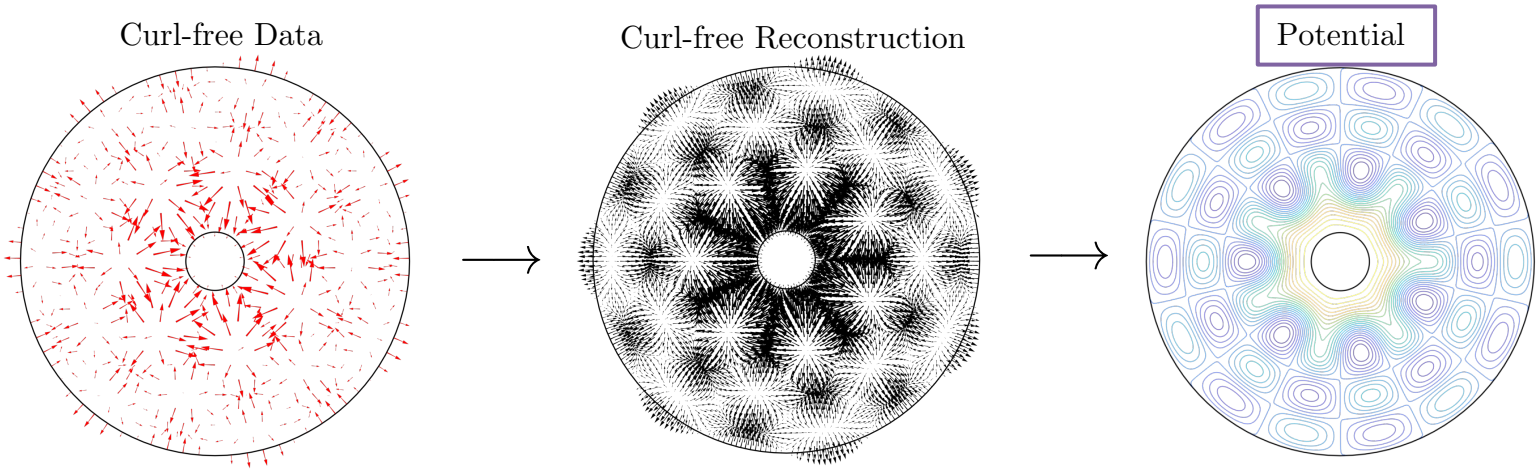
Property Preserving Kernels

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Boise State University



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Curl-free Kernel

References: Amodei & Benbourhim (1991), Narcowich & Ward (1994), Dodut & Rabut (2004), Fuselier (2006)

RBF-type solution

Consider the following *matrix-valued kernel* $\Phi : \mathbb{R}^d \times \mathbb{R}^d \longrightarrow \mathbb{R}^{d\times d}$

$$\Phi_{\text{cf}}(\mathbf{x},\mathbf{y}) = -\nabla\nabla^T \phi\left(\|\mathbf{x}-\mathbf{y}\|\right)$$

$\phi : [0,\infty) \to \mathbb{R}$ is a radial kernel

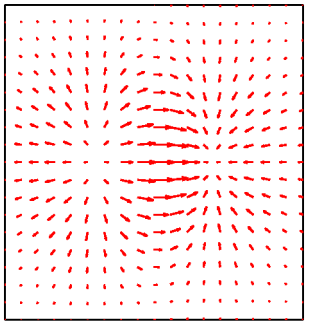
$\mathbf{y}$ is the "shift"

Claim: The columns of Φ_{cf} are curl-free

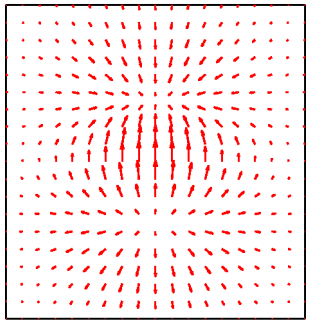
$$\implies \Phi_{\text{cf}}(\mathbf{x},\mathbf{y})\mathbf{e}_k = \nabla(\underbrace{-\nabla^T\phi(\|\mathbf{x}-\mathbf{y}\|)\mathbf{e}_k}_{g(\mathbf{x})}) = \nabla(g(\mathbf{x}))$$

Visualization d=2

$\Phi_{\text{cf}} \begin{bmatrix}1\\0\end{bmatrix}$



$\Phi_{\text{cf}} \begin{bmatrix}0\\1\end{bmatrix}$



Can show $\mathbf{\Phi}_{\text{cf}}$ is positive definite if ϕ is positive definite

Curl-free Kernel

References: Amodei & Benbourhim (1991), Narcowich & Ward (1994), Dodut & Rabut (2004), Fuselier (2006)

RBF-type solution

Consider the following *matrix-valued kernel* $\Phi : \mathbb{R}^d \times \mathbb{R}^d \longrightarrow \mathbb{R}^{d\times d}$

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$\phi : [0,\infty) \to \mathbb{R}$ is a radial kernel

$\mathbf{y}$ is the "shift"

Curl-free kernel regression of $\left\{\mathbf{u}(\mathbf{x}_j)\right\}_{j=1}^N$

$$\mathbf{s}_{\text{cf}}(\mathbf{x}) = \sum_{j=1}^N \Phi_{\text{cf}}(\mathbf{x},\mathbf{x}_j)\mathbf{c}_j, \quad \mathbf{s}_{\text{cf}}(\mathbf{x}_k) = \mathbf{u}_k, \ k = 1,\dots,N, \quad \mathbf{c}_j \in \mathbb{R}^d$$

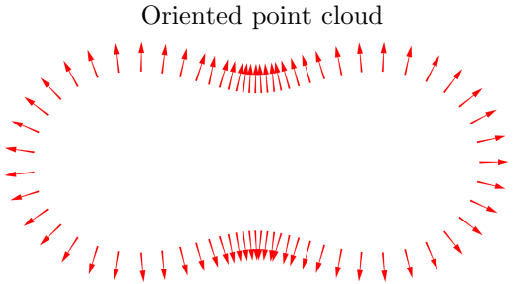
Bonus: a scalar potential for the field can also be obtained for free (Fuselier & W (2017))

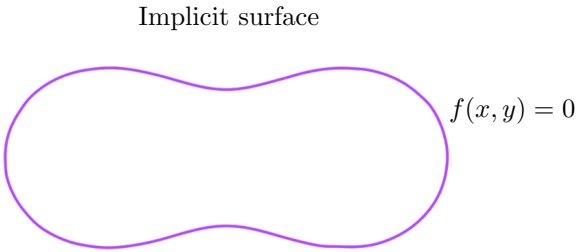
$$\mathbf{s}_{\text{cf}}(\mathbf{x}) = \sum_{j=1}^N \Phi_{\text{cf}}(\mathbf{x},\mathbf{x}_j)\mathbf{c}_j = \nabla\left(-\sum_{j=1}^N \nabla^T\phi\left(\|\mathbf{x}-\mathbf{x}_j\|\right)\mathbf{c}_j\right) = \nabla(s(\mathbf{x}))$$

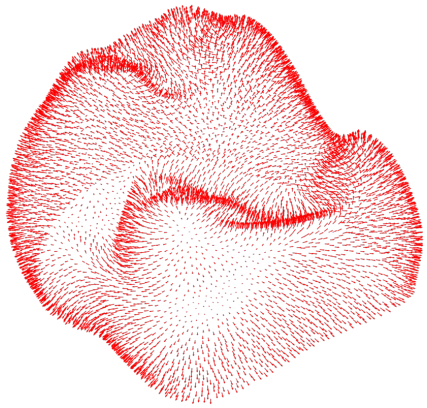
Don't have to solve a Poisson problem to recover the potential

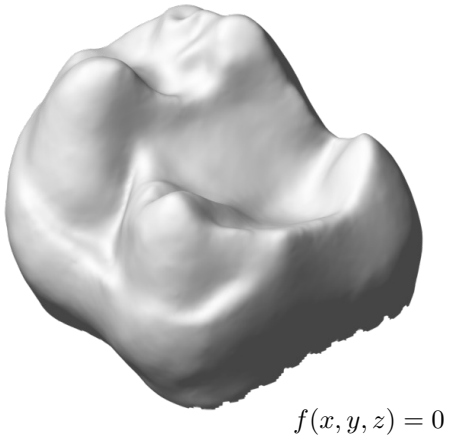
Application: Implicit surface reconstruction

Problem: Given an oriented point cloud $\{(\mathbf{x}_j, \mathbf{n}_j)\}_{j=1}^{N}$, find a function f such that the level-set $S = \{\mathbf{x} \in \mathbb{R}^d \,|\, f(\mathbf{x}) = 0\}$ fits the point cloud. S is called the zero-level implicit surface.





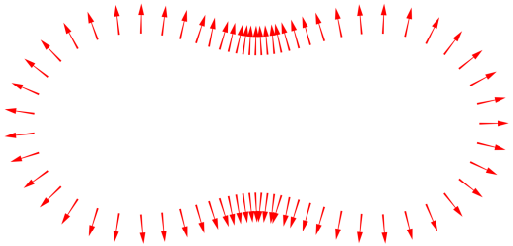




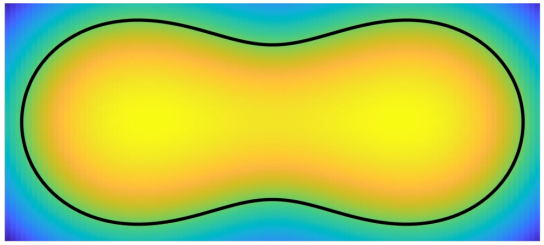
Proposition 1 *A vector field* $\mathbf{n}$ *is curl-free in* $\mathbb{R}^d$ *if and only if locally there exists a scalar potential* $f : \mathbb{R}^d \longrightarrow \mathbb{R}$ *such that* $\mathbf{n} = \nabla f$. *Furthermore,* f *is unique up to a constant.*

Reconstruction idea

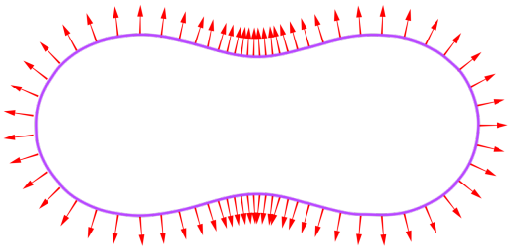
Input: oriented point cloud



Potential f



Output: implicit surface



$$X = \{\mathbf{x}_j\}_{j=1}^N \quad \{\mathbf{n}_j\}_{j=1}^N$$

Step 1: Solve for curl-free kernel expansion coefficients

$$\min_{\mathbf{c}\in\mathbb{R}^{2N}} \left[\sum_{j=1}^{N} \|\mathbf{s}_{\text{cf}}(\mathbf{x}_j) - \mathbf{n}_j\|^2 + \lambda\, \mathbf{c}^T K_{XX} \mathbf{c} \right]$$

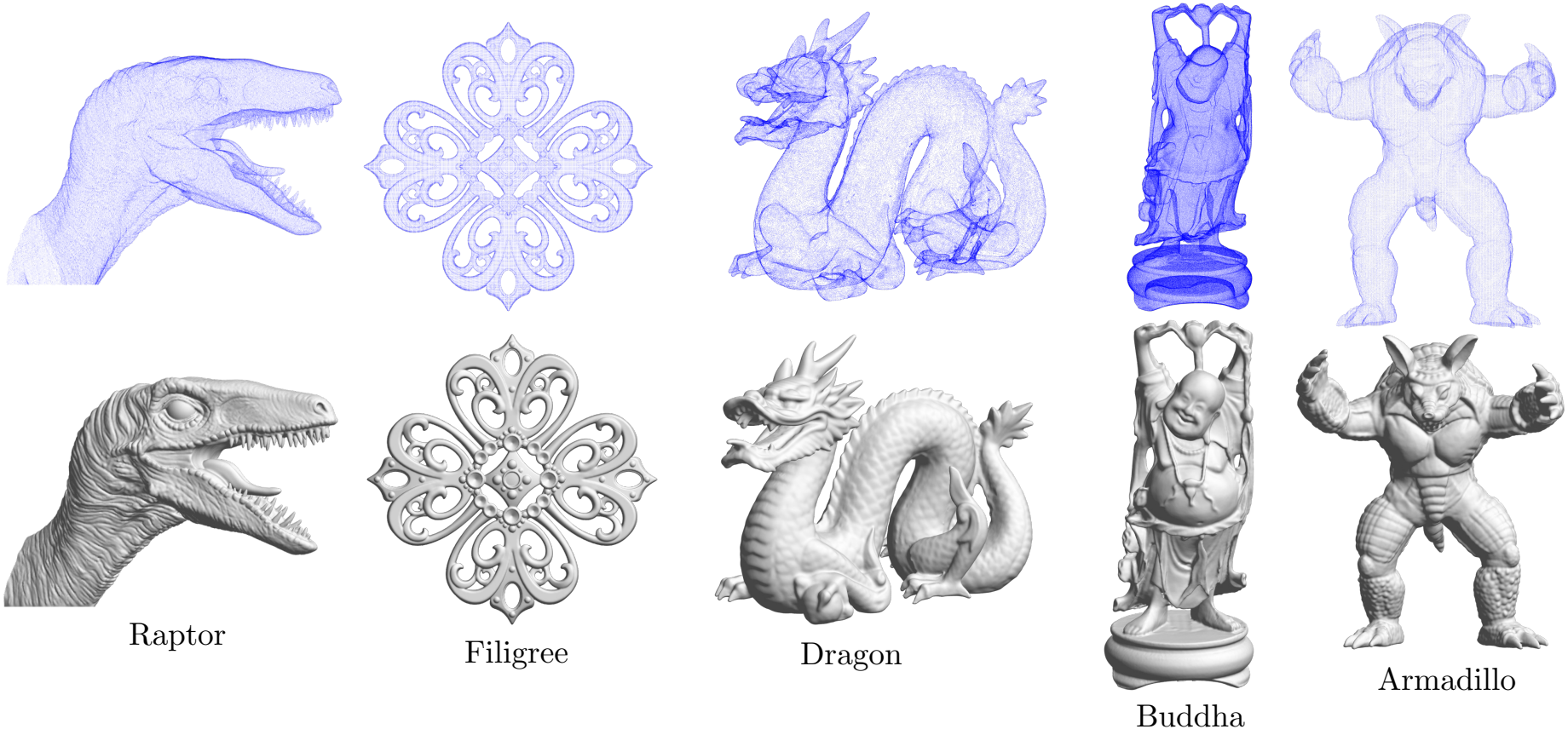
Step 2: Extract the potential for the field

$$s(\mathbf{x}) = -\sum_{j=1}^{N} \left[\nabla\phi\left(\|\mathbf{x} - \mathbf{x}_j\|\right)\right]^T \mathbf{c}_j$$

Step 3: Shift the potential so it's zero-level surface fits the point cloud

Step 4: Use isosurface extractor (e.g. marching cubes) to obtain the zero-level implicit surface

Results



Global method just presented to too computationally expensive for large N

Local methods developed based on partition-of-unity
Drake, Fuselier, & W (2022)

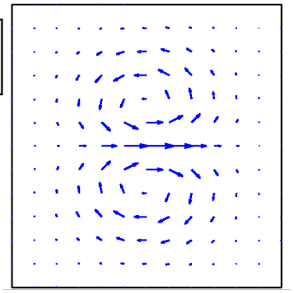


Need robust techniques for dealing with noisy normals and noisy sample points

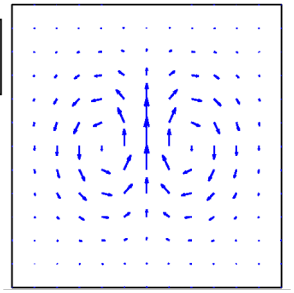
More customized kernels

Divergence-free kernels Φ_{df} can also be designed

$\Phi_{\text{df}} \begin{bmatrix} 1 \\ 0 \end{bmatrix}$



$\Phi_{\text{df}} \begin{bmatrix} 0 \\ 1 \end{bmatrix}$

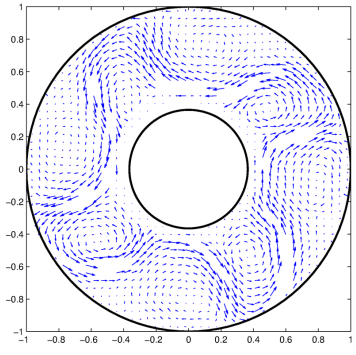


These can be combined with curl-free kernels for various applications

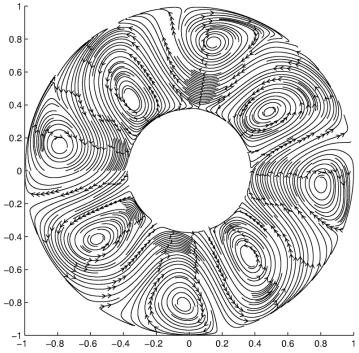
Narcowich, Ward & W (2007), Wendland (2009), Fuselier & W (2009), Schräder & Wendland (2011), Harlander et. al. (2014), Fuselier, Shankar & W (2016), Keim & Wendland (2016), Fuselier & W (2017), Fisher, Fasshauer & Gao (2024), Sharma et. al. (2026)

Example: Helmholtz-Hodge decomposition

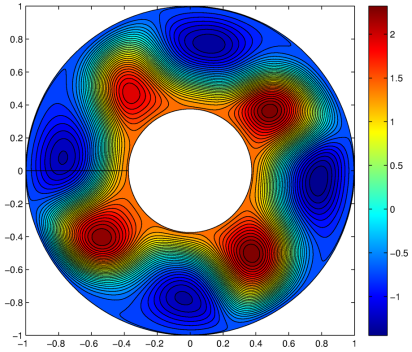
PIV Data



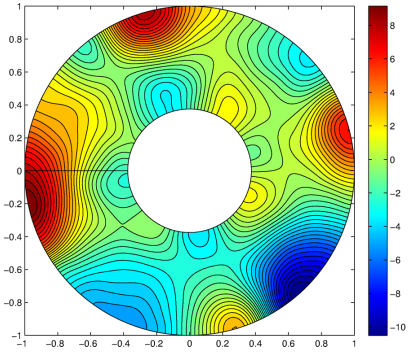
Reconstructed flow field



Divergence-free potential



Curl-free potential



Need techniques for incorporating uncertainty (and faster algorithms)